

Coffee Quality & Production Analysis Using SQL

1. Introduction

Coffee quality plays a crucial role in the global coffee industry. Companies need to identify high-quality coffee-producing regions and understand production trends to improve sourcing strategies. This project uses SQL to analyze coffee quality and production data.

2. Objective

The main objectives of this project are:

- Analyze coffee quality scores.
- Identify top-performing countries.
- Evaluate coffee varieties.
- Analyze processing methods.
- Study production trends.
- Generate business insights.

3. Dataset Description

The dataset contains 100 coffee samples collected from multiple countries.

Columns included:

- ID
- Country
- Variety
- Processing Method
- Bags Produced
- Total Score

4. Database Design

A relational database was created in MySQL.

Table Name: coffee

Primary Key:

- id

5. SQL Analysis Performed

Basic Analysis

- Total Records
- Distinct Countries
- Average Score

Aggregate Analysis

- Maximum Score
- Minimum Score
- Average Quality
- Total Production

Business Analysis

- Country Ranking
- Variety Performance
- Processing Method Analysis
- Production vs Quality

Advanced Analysis

- Window Functions
- Ranking Analysis
- Subqueries
- CASE Statements

6. Results

Key findings include:

- Kenya ranked highest in coffee quality.
- Ethiopia showed consistent quality performance.
- Brazil produced the highest volume.
- Washed processing method delivered superior quality.
- Premium varieties achieved higher scores.

7. Business Recommendations

- Increase sourcing from Kenya and Ethiopia.
- Prioritize Washed processing methods.
- Expand partnerships with high-performing producers.
- Monitor quality metrics alongside production volume.

8. Conclusion

This project demonstrates how SQL can be used to transform raw coffee production data into actionable business insights. The analysis supports strategic sourcing decisions and helps businesses optimize coffee quality and production efficiency.